REACTION PROFILES

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Potential energy diagrams are very useful in studying energy changes of reactions. They show:

- energy levels of reactants
- energy levels of products
- activation energy
- enthalpy change for the reaction.

Energy must be added to start a reaction. This is because an **energy barrier** (e.g. to break bonds) must be overcome. An energy barrier separates substances used in a reaction (reactants) from the products created. Overcoming the barrier to enable one substance to be converted into another requires the input of energy. This energy input is called the **Activation Energy** (E_a). Activation energy is the minimum amount of energy required to allow a reaction to occur; for example, the head of a match will burst into flame when it is struck – the heat from the friction starts the reaction.

1 Name another reaction that requires energy to start it.

For some reactions the energy available at room temperature is enough to overcome the activation energy. These reactions are said to be **spontaneous**. Iron rusting is a spontaneous reaction.

2 Define *spontaneous*.

3 Describe one other spontaneous reaction.

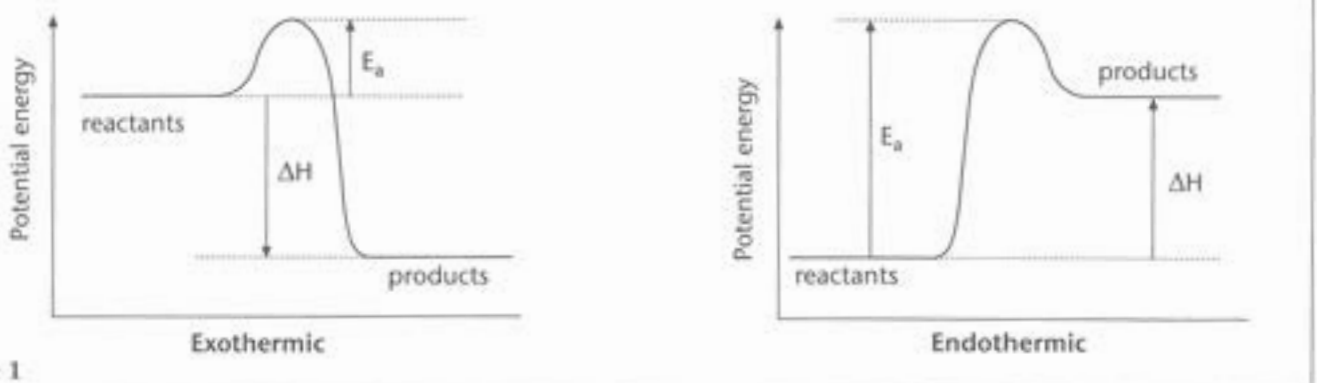


Figure 1

Figure 1 has two potential energy diagrams; one showing an exothermic reaction and the other showing an endothermic reaction.

4 Use figure 1 to help you answer the following.

a In the exothermic reaction, which is at the higher energy level: products or reactants?

b In the exothermic reaction, has energy been taken in or released by the reaction?

c In the exothermic reaction, is ΔH positive or negative?

d In the endothermic reaction, which is at the higher energy level: products or reactants?

e In the endothermic reaction, has energy been taken in or released by the reaction?

f In the endothermic reaction, is ΔH positive or negative?

5 Which is more likely to be a spontaneous reaction at room temperature: a reaction with a high activation energy or a reaction with a low activation energy? Explain your answer.

Standard conditions

All values quoted in references such as textbooks are usually referring to the reaction carried out under standard conditions. Two specific terms are **Standard Laboratory Conditions (SLC)** and **Standard Temperature and Pressure (STP)**. Common temperatures used as standards are 0°C (273K) and 25°C (298K) and generally pressures are around 101.3 kPa (1 atmosphere).

6 How do you convert degrees Celsius (°C) to Kelvins (K)?

7 *The minimum amount of energy required to convert a stable system into a reactive system.* What does this define?

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8 Mark each statement as true or false.

a Activation energy is energy given out in a reaction.

b In chemical reactions, only bond-making takes place.

c ΔH is the shorthand way of representing enthalpy change.

d Energy is required to start all chemical reactions.

e The temperature 273 Celsius (°C) is the same as 0 Kelvin (K)

Catalysts

Catalysts are often used in industry. A catalyst acts by altering the path of the reaction and changing the activation energy required. Note that activation energy is **always** positive. Figure 2 shows the effect of a catalyst on a reaction. The solid lines represent the uncatalysed reaction and the broken lines represent the catalysed reaction.

A **catalyst** alters the rate of reaction (usually speeding it up) or allows it to proceed at a lower temperature. Catalysts are not used up during a reaction and are unchanged at the end of the reaction; therefore, they are 'recycled'.

9 Copy figure 2. On the graph, draw and label each of the following.

a The activation energy for the uncatalysed reaction

b The activation energy for the catalysed reaction

c The enthalpy change (ΔH) for the reaction.

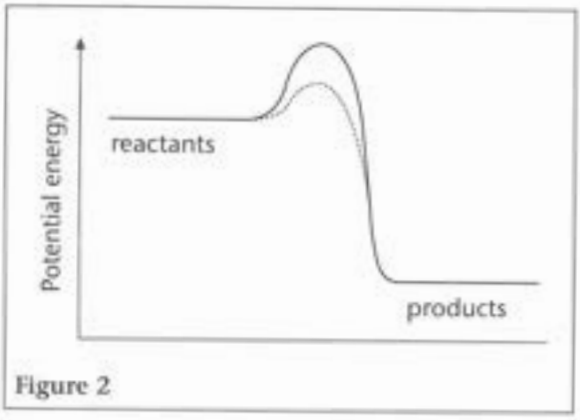


Figure 2

10 Is the reaction shown in figure 2 endothermic or exothermic? Explain your answer.

11 A student added manganese dioxide (a black insoluble solid) to hydrogen peroxide (a clear colourless liquid). Oxygen was given off and there seemed to be some black solid remaining. Describe how the student could test whether the manganese dioxide catalysed the reaction or whether it was a reactant itself.

12 What are the standard conditions under which heats of reactions are usually quoted?

Catalysts can also allow reactions to take place under milder (e.g. less heat and pressure) conditions, which saves energy. This doesn't give more products than the uncatalysed reaction does – just the same amount, but more quickly. Many transition metals and their compounds are catalysts.

Enzymes

Enzymes can be thought of as **biological catalysts**. They are much more specific than inorganic catalysts and only work at specific temperatures and pH levels. Enzymes act by forming temporary bonds with the molecules that are involved in a chemical reaction. They are usually made of complex proteins and can be **denatured** (their properties destroyed) at high temperatures. Enzyme names often end in '-ase' and refer to the reaction they catalyse. For example, lipases are enzymes that split up lipids (fats and oils).

Life as we know it would not exist without enzymes. Enzymes allow metabolic processes to react many times faster than would normally be possible. Enzyme reactions also occur at much lower temperatures than would otherwise be the case. This is important in cells because without enzymes the heat energy required for many reactions would be too high for the cell to survive. Enzymes are also used in processes such as making bread, beer and cheese.

13 Outline how catalysts work to speed up reactions.

14 Describe a similarity between enzymes and inorganic catalysts.

15 Describe a difference between enzymes and inorganic catalysts.

16 Give two reasons why a manufacturer might use catalysts in industrial reactions.

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